Dingjin Chu (初鼎晋)

Ph.D. Student, AGH 282
Division of Agricultural Sciences and Natural Resources
Natural Resource Ecology and Management
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Research Interests

Forest Functions, Forest structures, Forest disturbances, urban ecology
Remote Sensing, Forest inventory, Process-based model, Machine learning

Educations

Ph.D. Student, Quantitative Ecology, Oklahoma State University, Stillwater, Oklahoma, United States, May 2025 – Present (4.0 cumulative GPA)
Advisor: Lu Zhai (Asst. Prof.)

M.Sc., Forestry Ecology, Beijing Forestry University, China, 2020 - June 2023
Advisor: Kangning He (Prof.)

- Assess Forest functions across various stand structures
- Explore the relationship between timber production and forest productivity;
- Identify feasible indices derived from Sentinel-2 data to distinguish different vegetation types
- Thesis: Assessing the Forest States in the Eastern Part of Qilian Mountain Using Tree Rings and MODIS

B.Sc. (Agr.), Soil & Water Conservation and Desertification Combating, Shenyang Agricultural University, China 2016 - 2020

Publications

Chu, D., He, K., Lin, S., Zuo, Y, Chen, X. (2022). Comparison of forest and shrublands in potentially afforested areas based on remote sensing and field surveys. Acta Ecologica Sinica, 42(18), 7362-7371. (doi: 10.5846/ stxb202203210685)

Chu, D., Zhang, L., Wu, H., He, H., & Ren, X. (2026). Future climate extremes enhance interannual variability of net primary production in Central Asia. Environmental Research Letters. https://doi.org/10.1088/1748-9326/ae6dd4

Chen, X., Li, Y., Zuo, Y., Lin, S., **Chu, D**., He, K. (2022). Effects of stand characteristics on herbaceous diversity. *Acta Botanica Boreali-Occidentalia Sinica*, 42(08),1396-1407. (doi:10.7606/j.issn.1000-4025.2022.08.1396)

Wu, Han, Li Zhang, Honglin He, Xiaoli Ren, Mengyu Zhang, **Dingjin Chu**, and Yuanhao Zheng. (2025) "Enhanced interannual variability of the terrestrial carbon sink in China under high emissions." Environmental Research Letters

Research grants and rewards

2018-2019 Second class scholarship of Shenyang Agriculture University
2018-2019 Merit student award of Shenyang Agriculture University
2018-2019 Model student award of Shenyang Agriculture University
2020-2021 First class scholarship of Beijing Forestry University
2021-2022 First class scholarship of Beijing Forestry University
2024 National Olympic Sports Cup Adult Swimming Qualification Competition (8th)
2026 Afanasiev Distinguished Graduate Fellowship Award
2026 Third Place ($50), Student Poster Competition, GIS Day at OSU, Oklahoma State University
2026 Applied for NASA FINESST (Future Investigators in NASA Earth and Space Science and Technology) Fellowship

Conference Presentations

2026
10/2026 - Chu, D, **Local-scale bias in global products of forest aboveground biomass related to structural diversity, stand density, and forest age. Poster at GIS Day at OSU**, October 20, 2026, Stillwater, Oklahoma (Third Place, Student Poster Competition).
10/2026 - Chu, D, **Patterns and Drivers of Urban Forest Cooling Across U.S. Cities. Talk at the Society of American Foresters (SAF) National Convention**, October 6–9, 2026, Tacoma, Washington.
10/2026 - Chu, D, **Understanding the Drivers of Local-Scale Bias in Global Forest Aboveground Biomass Maps. Poster at the Society of American Foresters (SAF) National Convention**, October 6–9, 2026, Tacoma, Washington.
4/2026 - Chu, D, **The factors influencing the accuracy of global forest biomass map to estimate local carbon storage: A case study of Tribal Nations in the U.S. Talk at the Natural Resource Conference**, Oklahoma City.

2025
12/2025 - Chu, D, **Understanding the Drivers of Local-Scale Bias in Global Forest Aboveground Biomass Maps. Poster at the AGU, December 15-29, 2025 New Orleans.**

2023

8/2023 - **Chu, D.** Zhang, L., Wu, H., He, H., Ren, X. **Impacts of future climate extremes on interannual variability of terrestrial ecosystem productivity in Central Asia based on CMIP6**. Poster at conference Actual Issues of Environmental Research for Sustainable Development in Arid Zones, Issyk-Kul, Kyrgyzstan

PROFESSIONAL SERVICE Reviewer for Journals:

TBD

TEACHING

Teaching Assistant:

Spring 2027 NREM 3125 Forest Measurements; Teaching hours: 40

Guest Lecturer:

Training:

Fall/2026 – ITA Training - Language in the American Classroom

08/2026 – Graduate Teaching Assistant Conference; Training Hours: 8

08/2025 – Graduate Teaching Assistant Conference; Training Hours: 8

Volunteering & Outreach

10/2026 - Volunteer in the Walk in the Forest activity at Martin Park Nature Center, Oklahoma City, to share forestry knowledge with local visitors
4/2026 - Oklahoma Division SAF Spring Meeting; Filed trip to understand how wildfire influences forest and its recovery status

10/2025 - **Volunteer** in the Walk in the Forest activity to spread the forestry knowledge to The Local people at the Forest Heritage Center in Beavers Bend State Park, Oklahoma

9/2025 - Fire, Timber, Wildlife, & Cattle Field Tour at ODWC Pushmataha WMA , 2025
Reach out the landowner to manage lands using prescribed fire

Professional and research experience

Department of Natural Resource Ecology and Management, Oklahoma State University

Graduate Research Assistant | June 2025 – Present

- Estimate forest aboveground biomass using LiDAR data.
- Assess the cooling effects of urban green spaces.
- Investigate forest stability in response to climate extremes.

Institute of Geographic Sciences and Natural Resources Research, Chinese Academy of Sciences

Research Assistant | July 2023 – May 2025

- Distinguished global forest disturbances, including forest degradation, deforestation, and regrowth.
- Quantified the contributions of climate variability and extremes to vegetation productivity in semi-arid regions.
- Assimilated model parameters into process-based ecosystem models.
- Processed and analyzed large-scale geospatial datasets, including TRENDY and CMIP6 climate data.

Beijing Forestry University

Master's Student

Improving the Functions of Vegetation in the Qilian Mountains by Adjusting Vegetation Structures

- Conducted forest field investigations, assessing stand structural diversity, tree rings, herb diversity, soil properties, and litter characteristics.

Optimizing and Allocating Small Watershed Landscape Patterns

- Conducted vegetation surveys across plantations, natural forests, shrublands, and grasslands.
- Processed geospatial data, including Sentinel-2 imagery, topography, and climate datasets.
- Applied remote sensing data from Sentinel-2, MODIS, and Landsat.
- Used InVEST to assess soil erosion, carbon storage, and water yield.

Water Consumption Characteristics of Main Afforestation Tree Species in the Loess Area of the Huangshui River Basin and Its Water Environment Capacity

- Monitored soil moisture and tree sap flow.
- Operated field monitoring equipment, including WatchDog 2800, Sapflow32, and Vantage Pro2 Plus.

Community services

June 2025 – May 2026 **GSO Speaker Series Committee member**, Department of Natural Resource Ecology and Management, Oklahoma State University, Stillwater, Oklahoma, United States,
Harmonizing students and faculties to contact potential speakers and organize the activities.

Member, Society of American Foresters (SAF), Oklahoma State University, Stillwater, Oklahoma, United States
Member, American Geophysical Union (AGU)
Member, Society of American Foresters (SAF**)**

Skills

Computer skills: R (Proficient in tidyverse, tidymodels, terra, data.table, and other packages), Google Earth Engine (Download and process), Python, Linux (basic command for running CLM), QGIS, ArcGIS, ArcGIS pro
Language: English (IELTS:7); Chinese (native)
Field investigation: Conducted field surveys to analyze vegetation structure, assess soil properties, and study herb diversity in ecosystems including plantations, forests, shrublands, and grasslands. Collected tree rings for further analysis.
Remote sensing: Experience with Sentinel-2, MODIS, and Landsat data
Process-based models: CLM (Running), CMIP6 (Process data), Trendy (Process data)

Personal interests

Hiking, Swimming, Cycling, Climbing
Music, Movies, Reading